Project Report: Explainable AI (XAI) Framework for Crop Area Estimation & Survey Validation

1. Executive Summary

This project aims to develop a transparent, trust-based AI system for large-scale crop area estimation. While traditional Deep Learning models (like LSTMs) achieve high accuracy in crop classification, they function as "black boxes," failing to explain *why* a specific pixel was classified as a certain crop. This lack of transparency limits adoption by government bodies and insurance agencies.

Our solution integrates **Sentinel-1 (Radar)** and **Sentinel-2 (Optical)** satellite data into a time-series **LSTM model**, augmented with an **Explainable AI (SHAP)** module. This system not only classifies crops but also validates its decisions by cross-referencing with administrative GIS boundaries (Districts/Tehsils) to detect discrepancies in official government agricultural surveys.

2. Problem Statement

- **The Trust Gap:** Governments rely on manual surveys (Giridawari) which are slow and prone to human error. Satellite-based AI models are fast but opaque; officials cannot verify if the model is learning real phenology or just fitting noise.
- **The Cloud Problem:** Optical satellites (Sentinel-2) are blind during the Kharif (monsoon) season, leading to data gaps.
- **The Solution:** A fused Radar-Optical model that works in all weather, equipped with an "Explanation Engine" that proves the model's physical validity.

3. System Architecture

The system is divided into four technical modules:

Module A: Data Ingestion & Pre-Processing

- **Objective:** Create a clean, continuous time-series for every pixel.
- **Inputs:**
    - **Sentinel-1 (SAR):** VV and VH polarizations (Backscatter coefficient σ^0). *Crucial for seeing through monsoon clouds.*
    - **Sentinel-2 (Optical):** Band 4 (Red) and Band 8 (NIR) to calculate **NDVI** (Normalized Difference Vegetation Index).

- **Novelty Technique (Temporal Interpolation):** Since satellites do not revisit daily, we use **Linear Interpolation** to fill gaps, creating a standard tensor of shape (Time Steps, Features).
    - *Input Tensor Shape:* [12 Months × 30 Days, 3 Features (NDVI, VV, VH)]

Module B: The Deep Learning Core (LSTM)

- **Objective:** Classify the crop type based on growth patterns over time.
- **Model: Long Short-Term Memory (LSTM)** Network.
- **Why LSTM?** Unlike standard CNNs that look at a static image, LSTMs analyze the *sequence*. They understand that "Low Greenness in May" followed by "High Greenness in August" signifies a specific crop cycle (e.g., Rice).

Module C: The Explainability Engine (SHAP)

- **Objective:** Open the "Black Box."
- **Technique: SHAP (SHapley Additive exPlanations)**.
- **Output:** A "Feature Importance" graph for every prediction.
    - *Example Output:* "Classified as Wheat because Feature NDVI was High in February."
    - *Validation:* We check if this matches the biological reality of Wheat (which indeed peaks in Feb).

Module D: Survey Validation (GIS Aggregation)

- **Objective:** Compare AI results with Government Data.
- **Process:**
    1. Load Government GIS Boundaries (Shapefiles for Districts).
    2. Count "AI-predicted Rice Pixels" inside the boundary.
    3. Compare with "Official Govt Reported Area."
    4. Flag districts with high discrepancy (>15%) for manual audit.

4. Dataset Details

Primary Dataset: CropHarvest (NASA)

- **Description:** A global dataset of labeled agricultural points, specifically designed for machine learning.
- **Data Availability:** Public (Open Source).
- **Why we use it:** It abstracts away the complex downloading of raw Terabytes of satellite data. It provides Python tools to fetch ready-to-use time-series arrays.
- **Link:** CropHarvest GitHub

Validation Dataset: Data{Meet} India Maps

- **Description:** Community-sourced spatial data for India.
- **Data Availability:** Public (Open Source).
- **Use Case:** Provides the .shp (Shapefiles) for Indian Districts and Tehsils to aggregate our pixel counts.
- **Link:** Data{Meet} Maps

Optional Pre-Training: BreizhCrops

- **Description:** A high-quality benchmark dataset from France.
- **Use Case:** If Indian labels are scarce, we pre-train the LSTM on this massive dataset to learn general crop features, then "fine-tune" on the smaller Indian dataset.

5. Execution Roadmap (Step-by-Step)

Phase 1: Setup & Exploration (Week 1)

- **Goal:** Get the code running.
- **Action:**
    - Install cropharvest and shap libraries.
    - Download the "Kenya" or "Togo" subset from CropHarvest (closest to Indian agriculture in terms of smallholder farms).
    - Visualize the time-series: Plot NDVI vs. Time for 10 Maize samples and 10 Non-Maize samples.

Phase 2: Model Development (Weeks 2-3)

- **Goal:** Build the LSTM.
- **Action:**
    - Build a PyTorch model with:
        - Input Dimension: 3 (VV, VH, NDVI)
        - Hidden Layer: 64 Units
        - Output: Number of Crop Classes
    - Train on 80% of data, Test on 20%.
    - Target Accuracy: >85%.

Phase 3: Integration of XAI (Week 4)

- **Goal:** Apply SHAP.
- **Action:**
    - Pass the trained model into shap.DeepExplainer.
    - Generate **Force Plots** for misclassified examples to understand *why* the model failed (e.g., "Did it confuse clouds for a crop?").

Phase 4: GIS Validation (Week 5)

- **Goal:** The "Government" simulation.
- **Action:**
    - Take a specific region.
    - Sum the pixel areas.
    - Create a simple Matplotlib map showing "Discrepancy Hotspots.

6. Tools & Technologies

Component	Tool / Library	Purpose
Language	Python 3.9+	Core programming.
Deep Learning	PyTorch or TensorFlow	Building the LSTM model.
Data Access	cropharvest	Downloading satellite time-series.
Explainability	shap	Generating explanation graphs.
Geospatial	geopandas & rasterio	Handling maps and shapefiles.
Environment	Google Colab	Free GPU access for training.

7. Expected Outcomes (Deliverables)

1. **The Algorithm:** A trained LSTM model capable of classifying crops using dual-sensor data.
2. **The Evidence:** SHAP plots proving the model is using physical parameters (phenology) rather than noise.
3. **The Report:** A comparative analysis showing where Government Survey data might be inaccurate based on AI evidence.

This architecture ensures you have **high scientific novelty** (via the XAI/SHAP component) while remaining **technically feasible** for a student project by using ready-made data loaders like cropharvest.